Somjit Roy

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Research Interests and Skill Set

My research advances next-generation statistical modeling frameworks that bridge data-driven learning with scientific reasoning. I develop novel methodologies for complex real-world problems in materials discovery, geophysics, bioinformatics, and finance. Working at the intersection of scientific machine learning and Bayesian modeling & computation, I focus on scientifically guided probabilistic inference, physics-informed modeling, approximate Bayes, tree-based models, and Bayesian optimization, with the goal of building interpretable, scalable, and scientifically grounded learning systems.

Core methods: Probabilistic SciML | Symbolic Regression | Approximate Bayes | Bayesian Optimization

Education and Training

Texas A&M University, College Station, TX, USA — 2023–present

Doctor of Philosophy (Ph.D.) in Statistics — **GPA: 4.00/4.00**

Advisors: **Bani K. Mallick** (TAMU Statistics) & **Debdeep Pati** (UW–Madison Statistics)

Area of Study: Probabilistic Scientific Machine Learning and Computational Bayes

University of Calcutta, Kolkata, West Bengal, India — 2021–2023

Master of Science (M.Sc.) in Statistics, Ranked First Class First — **GPA: 8.15/10.00**

St. Xavier's College (Autonomous), Kolkata, West Bengal, India — 2018–2021

Bachelor of Science (B.Sc.) in Statistics, Honours — **GPA: 8.63/10.00**

Experience

Upstart Network, Inc. — June 2026–Aug 2026

Research Scientist Intern, Machine Learning — Remote

- Developed a **selection-bias-aware NPV-RMSE** for funded borrowers, revealing 9−11% higher applicant-level error than historical estimates.
- Designed two-stage transport weighting, improving applicant covariate balance by 36%.
- Built propensity models using `AWS` and machine learning architectures in `Python` across 5.4M+ loans and 5K+ features.

Los Alamos National Laboratory — May 2025–Aug 2025

Graduate Summer Intern — Los Alamos, NM, USA

- Developed a scalable **physics-informed neural operator** for acoustic-wave modeling in complex, heterogeneous media.
- Scaled training with `PyTorch DDP`, `CUDA`, and multi-`GPU` high-performance computing.
- Generalized wavefield predictions to unseen spatial and temporal coordinates in held-out heterogeneous media.

Tata Electronics Pvt. Ltd. — May 2021–Aug 2021

Data Science Intern — Bengaluru, KA, India

- Analyzed **sandblasting and anodizing trials** to identify process drivers of gloss and texture and recommend in-spec parameter settings.
- Applied random Forest, support vector machines, ridge, and lasso regression to address multicollinearity and rank process parameters.
- Identified band and jig positions as key sources of variation, reducing production time by two person-days.

Papers and Manuscripts

Tags: ★ Probabilistic SciML ☆ Symbolic Regression ☆ Variational Inference ☆ Bayesian Optimization ☆ Quantum Cryptography

6. *Roy, S.*, Dey, P., & Mallick, B. K. (2026). VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees. (*Accepted at UAI 2026*). **arXiv** ↗ ★ ☆ ☆

5. *Roy, S.*, Jaiswal, P., Bhattacharya, A., Pati, D., & Mallick, B. K. (2026+). On Frequentist Regret Analysis of Fractional Gaussian Process Thompson Sampling. (*Submitted*). **arXiv** ↗ ☆

4. *Roy, S.*, Dey, P., Mallick, B. K., & Pati, D. (2026+). Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests. (*Submitted*). **arXiv** ↗ [2026 IMS Hannan Graduate Student Travel Award; 2nd place at Best of Statistical Science (BOSS) 2026 Conference (TidBit session); 2026 ASA SBSS Student Paper Award; selected for refereed oral presentation at ASA's SDSS 2026; featured by the Texas A&M College of Arts and Sciences (news ↗)]. ★ ✩

3. *Roy, S.*, Dey, P., Pati, D., & Mallick, B. K. (2026+). A Generalized Tangent Approximation based Variational Inference Framework for Strongly Super-Gaussian Likelihoods. (*Revision submitted: JASA Theory & Methods*). **arXiv** ↗ ✩

2. Kumar, A., Maitra, S., & *Roy, S.* (2024). Almost Perfect Mutually Unbiased Bases that are Sparse. *Journal of Statistical Theory and Practice*, **18**, 61. **doi** ↗ ✩

1. Chaudhury, S., Kumar, A., Maitra, S., *Roy, S.*, & Sen Gupta, S. (2022). A Heuristic Framework to Search for Approximate Mutually Unbiased Bases. In *Cyber Security, Cryptology, and Machine Learning* (CSCML 2022), Lecture Notes in Computer Science, vol. **13301**. Springer, Cham. **doi** ↗ ✩

Open-Source Software

GitHub `VaSST` (*Roy, S.* & Dey, P., 2026); `BayeSymX` (*Roy, S.* & Dey, P., 2026); `TAVIE-SSG` (*Roy, S.* & Dey, P., 2025)

CRAN `bayesestdft` (*Roy, S.* & Lee, S. Y., 2025); `GoodFitSBM` (Ghosh, S., *Roy, S.*, & Pati, D., 2024)

Technical Skills

Programming	Python, C++, R, Java
ML & Scientific Computing	PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, SciPy
Probabilistic Computing	PyMC, Stan/RStan, mcmc, coda
GPU & Distributed Computing	CUDA, GPU computing, PyTorch DDP, MPI, OpenMP
Systems, Cloud & Tools	Linux, AWS, Git, Hugging Face, Docker, LaTeX

Awards and Fellowships

2026 **IMS Hannan Graduate Student Travel Award**, Institute of Mathematical Statistics

2026 **SBSS Student Paper Award**, American Statistical Association

2026 **SDSS Student and Early Career Scholarship**, American Statistical Association

2026 **Best of Statistical Science (BOSS) TidBit Award**, Texas A&M University

2026 **Graduate Research and Presentation (RAP) Travel Award**, Texas A&M University

2024 **NSF Travel Grant**, for presenting at IISA 2024

2024 **Targeted Proposal Teams (TPT) Grant**, Research Assistantship | Texas A&M University

2024 **Selected for CMS^3-FAST Summer School**, Texas A&M University

2022 **R. C. Bose Memorial Book Prize**, Calcutta Statistical Association

2022 **IASc-INSA-NASI Science Academies Summer Research Fellowship**, Indian Academy of Sciences

2022 **OPHI Summer School Grant**, University of Oxford

2022 **IAOS Conference and Travel Grant**, World Bank

Talks and Presentations

2026 **Joint Statistical Meetings** Invited SBSS Award Presentation
Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests — Boston, MA

2026 **Symposium on Data Science & Statistics** Refereed Invited Talk
Hierarchical Bayesian Operator-Induced Symbolic Regression Trees for Structural Learning of Scientific Expressions — Milwaukee, WI

2025 **STAT CAFÉ** Seminar Talk
Bayesian Symbolic Trees for Structural Learning of Scientific Expressions — College Station, TX

2025 **SIAM Texas-Louisiana Sectional Meeting** Invited Talk
Physics-Informed Neural Operators for Seismic Wave Modeling — Austin, TX

2025 **Los Alamos National Laboratory Student Symposium** Poster
Physics-Informed Neural Operators for Seismic Wave Modeling — Los Alamos, NM

2024 **International Indian Statistical Association Conference** Contributed Talk
Tangent Approximation for Variational Inference in Different Exponential Families — Kochi, India

Leadership and Professional Activities

- **Workflow Workshop Organizer** **2024–2025**
  Statistics Graduate Student Association, Texas A&M University
  Led and coordinated graduate workshops on research workflow, computational practice, and academic productivity.
- **Organizing Committee Member** **2019–2020**
  Department of Statistics, St. Xavier's College (Auotnomous), Kolkata
  Served on the organizing committee for the annual departmental fest ϵpsilon δelta, supporting event planning and coordination.

References

Bani K. Mallick
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